UNIT-4

SEQUENTIAL SYSTEMS

SEQUENTIAL SYSTEMS: Digital system in which output at any time depends not only on the input at that time but also on previous inputs.

$$z(t) = F(x(0, t))$$

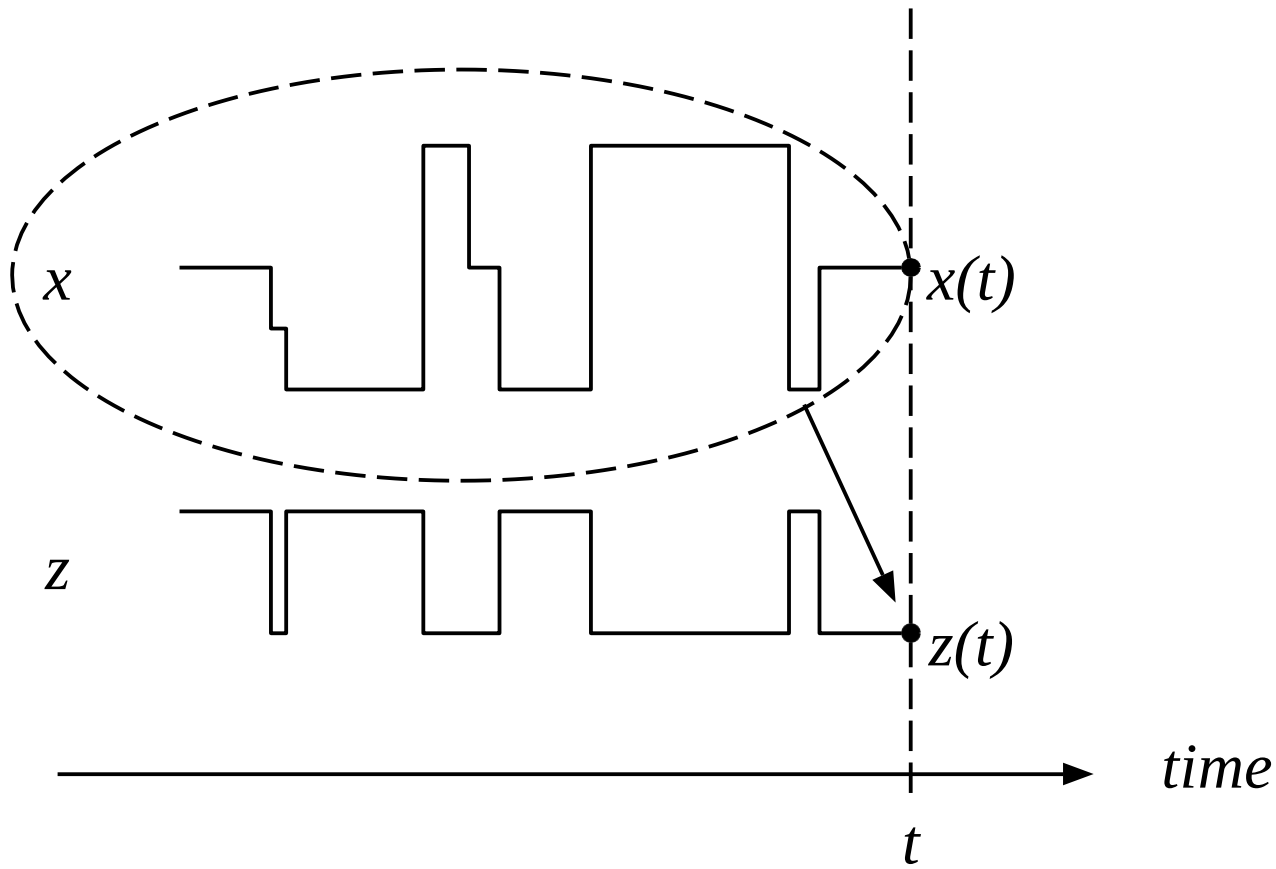


Figure 7.1: INPUT AND OUTPUT TIME FUNCTIONS.

SYNCHRONOUS AND ASYNCHRONOUS SYSTEMS

A *synchronous* sequential circuit is a system whose behavior can be defined at discrete instants of time.

The behavior of an *asynchronous* sequential circuit depends upon the input signals at any instant of time (that is time variable is continuous) *and* the order in which the inputs change.

SYNCHRONOUS AND ASYNCHRONOUS SYSTEMS

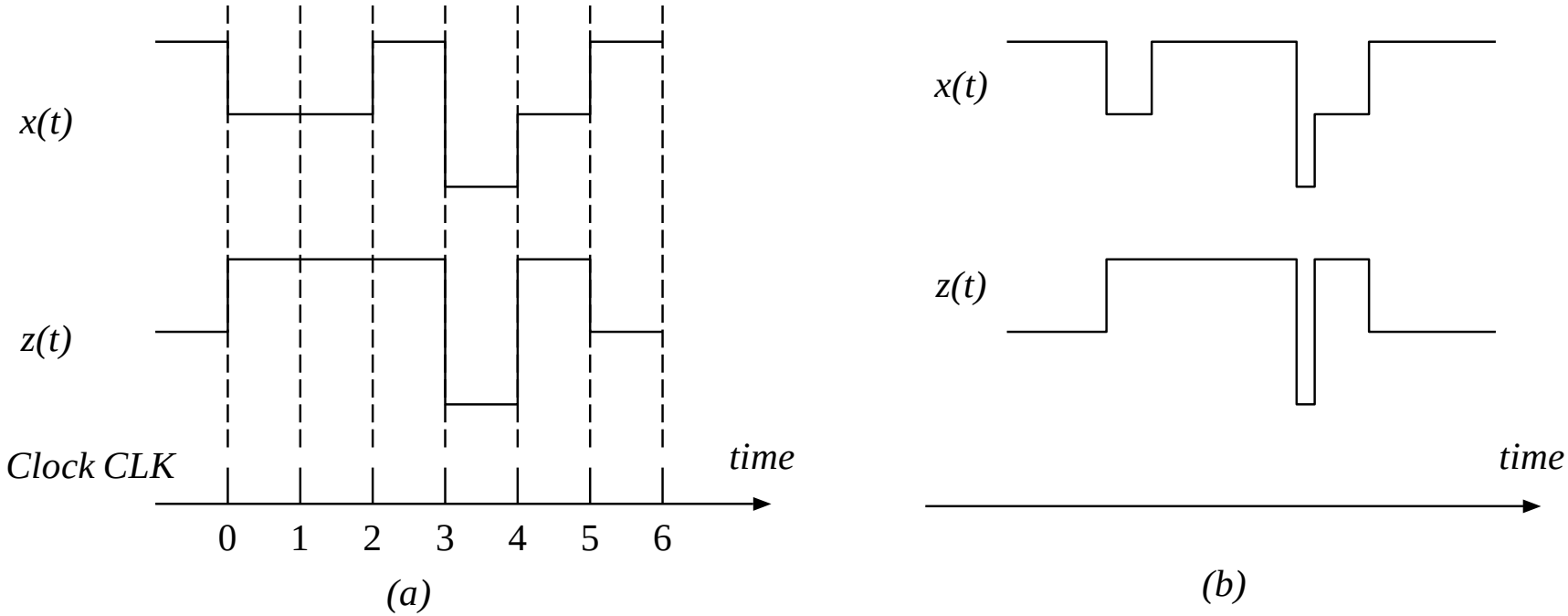


Figure 7.2: a) SYNCHRONOUS BEHAVIOR. b) ASYNCHRONOUS BEHAVIOR.

- CLOCK (clock cycle)
- I/O SEQUENCE $x(t_1, t_2)$: this sequence has length $t_2 - t_1 + 1$
- $x(2, 5) = aabc$
- $z(2, 5) = 1021$

SPECIFICATION OF SEQUENTIAL SYSTEMS

- Problem statement
- High-level specification (Input, output, function)
- State diagram
- State description
- Binary-level specification

STATE DESCRIPTION OF FINITE STATE SYSTEM

- Finite state machine /finite state system
- State description: consists of the input set I, the output set O, the state set S, the I initial state s(0) and following two functions:

State-transition function $s(t+1) = G(s(t), x(t))$

Output function $z(t) = H(s(t), x(t))$

STATE DESCRIPTION

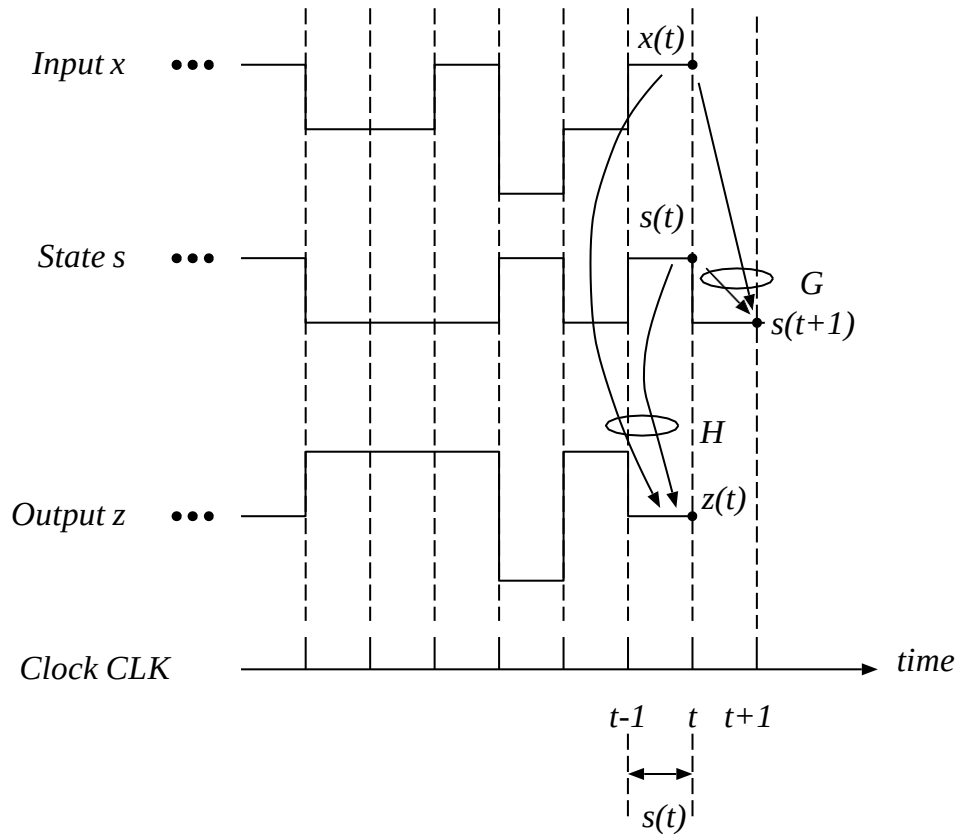


Figure 7.3: OUTPUT AND STATE TRANSITION FUNCTIONS

State-transition function $s(t+1) = G(s(t), x(t))$

Output function $z(t) = H(s(t), x(t))$

Example 7.1: SERIAL DECIMAL ADDER

x	1638753
y	3652425
Z	5291178

- LEAST-SIGNIFICANT DIGIT FIRST (at t=0)

t	0	1	2	3	4	5	6
x(t)	3	5	7	8	3	6	1
y(t)	5	2	4	2	5	6	3
z(t)	8	7	1	1	9	2	5

Example 7.3: STATE DESCRIPTION OF SERIAL ADDER

Input: $x(t), y(t) \in \{0, 1, \ldots, 9\}$
Output: $z(t) \in \{0, 1, \ldots, 9\}$
State: $s(t) \in \{0, 1\}$ (the carry)
Initial state: $s(0) = 0$

Functions: The transition and output functions are

$$s(t+1) = \begin{cases} 1 & \text{if } x(t) + y(t) + s(t) \geq 10 \\ 0 & \text{otherwise} \end{cases}$$

$$z(t) = (x(t) + y(t) + s(t)) \; mod \; 10$$

EXAMPLE:

t	0	1	2	3	4	5	6
x(t)	3	5	7	8	3	6	1
y(t)	5	2	4	2	5	6	3
s(t)	0	0	0	1	1	0	1
z(t)	8	7	1	1	9	2	5

Example 7.4: ODD/EVEN

Consider a system whose input has two values, called a and b, and whose output also has two values 1 and 0. The output at time t is 1 if number of b's in the input time function x(0,t) is even, and 0 otherwise.

TIME-BEHAVIOR SPECIFICATION:

Input: $x(t) \in \{a, b\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if } x(0, t) \text{ contains an even number of } b\text{'s} \\ 0 & \text{otherwise} \end{cases}$

I/O SEQUENCE:

t	0	1	2	3	4	5	6	7
x, z	$a, 1$	$b, 0$	$b, 1$	$a, 1$	$b, 0$	$a, 0$	$b, 1$	$a, 1$

Example 7.4: STATE DESCRIPTION OF ODD/EVEN

Input: $x(t) \in \{a, b\}$
Output: $z(t) \in \{0, 1\}$
State: $s(t) \in \{\text{EvEN}, \text{odd}\}$
Initial state: s(0)= EvEN

Functions: Transition and output functions

PS	$x(t) = a$	$x(t) = b$
EvEN	EvEN, 1	odd, 0
odd	odd, 0	EvEN, 1
	$NS,\ z(t)$	

MEALY AND MOORE MACHINES

Mealy machine

$$z(t) = H(s(t), x(t))$$

$$s(t+1) = G(s(t), x(t))$$

Moore machine

$$z(t) = H(s(t))$$

$$s(t+1) = G(s(t), x(t))$$

- EQUIVALENT IN CAPABILITIES

Example 7.5: MOORE SEQUENTIAL SYSTEM

Input: $x(t) \in \{a, b, c\}$
Output: $z(t) \in \{0, 1\}$
State: $s(t) \in \{S_0, S_1, S_2, S_3\}$
Initial state: $s(0) = S_0$

Functions: Transition and output functions:

PS	Input			
	a	b	c	
S_0	S_0	S_1	S_1	0
S_1	S_2	S_0	S_1	1
S_2	S_2	S_3	S_0	1
S_3	S_0	S_1	S_2	0
	NS			Output

REPRESENTATION OF STATE-TRANSITION AND OUTPUT FUNCTIONS

- STATE DIAGRAM

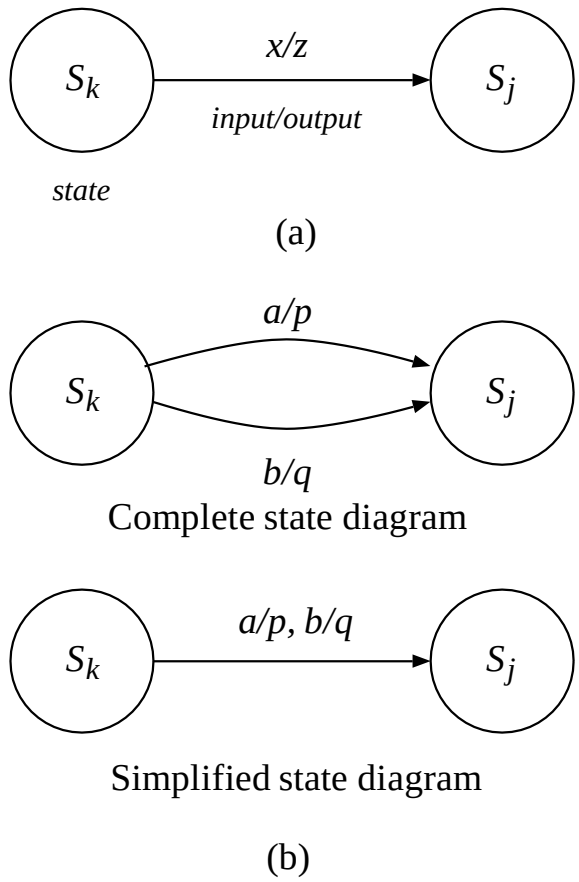


Figure 7.4: (a) STATE DIAGRAM REPRESENTATION. (b) SIMPLIFIED STATE DIAGRAM NOTATION.

Example 7.6

Functions: The transition and output functions are

$s(t)$	$x(t)$	
	a	b
S_0	S_1, p	S_2, q
S_1	S_1, p	S_0, p
S_2	S_1, p	S_2, p
	$s(t+1), z(t)$	

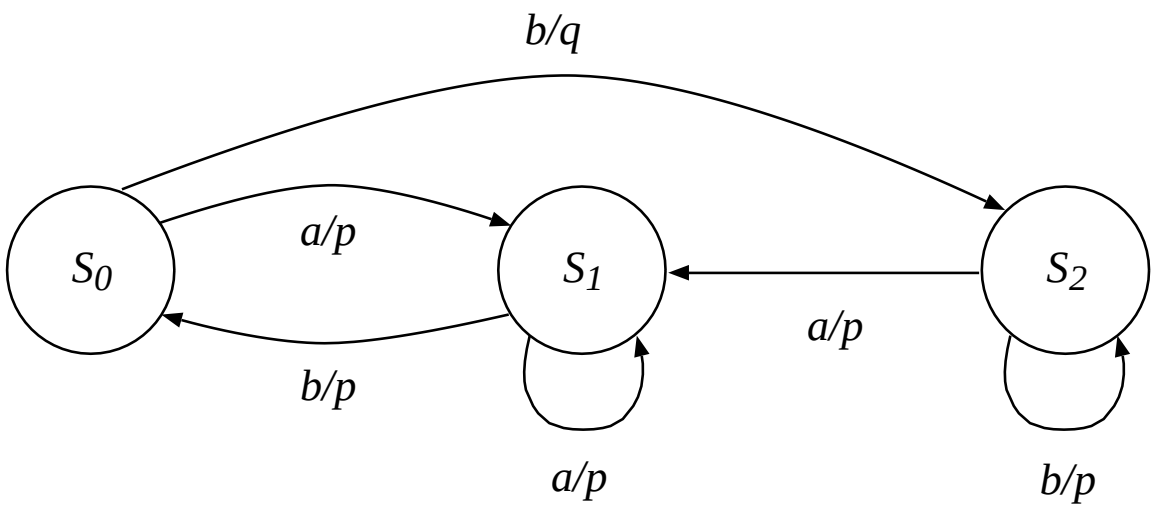


Figure 7.5: STATE DIAGRAM FOR EXAMPLE 7.6.

STATE DIAGRAM FOR A MOORE MACHINE

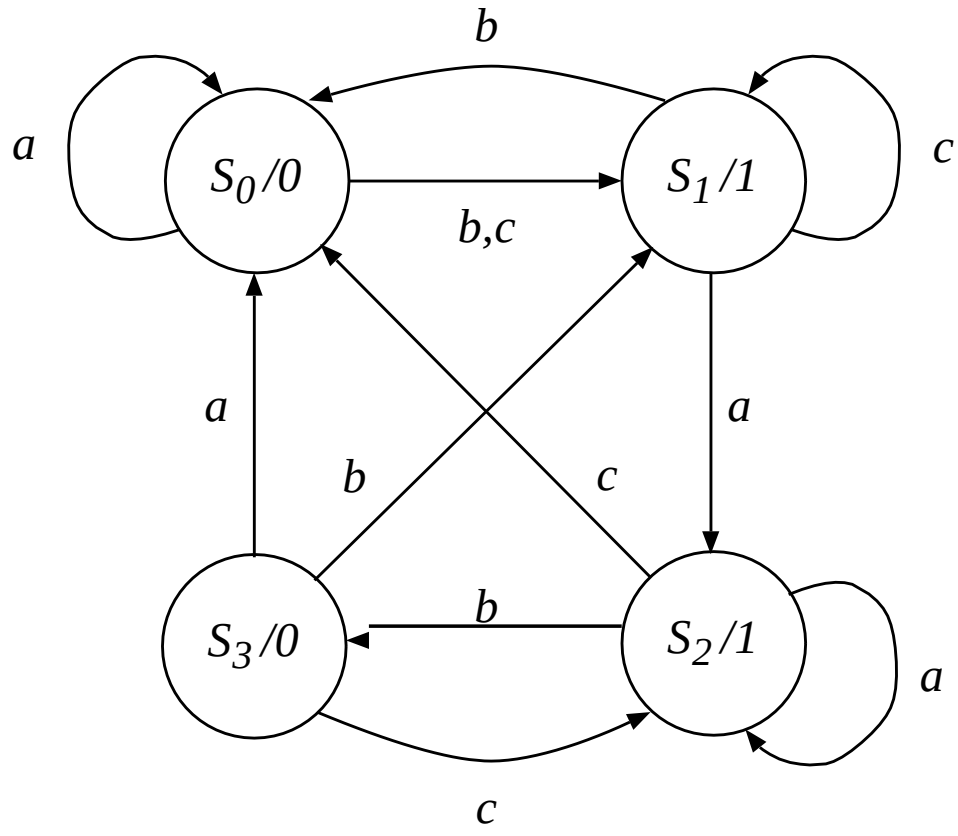


Figure 7.6: STATE DIAGRAM FOR EXAMPLE 7.5

Example 7.7: State diagram

Input: $x(t) \in \{0, 1, 2, 3\}$
Output: $z(t) \in \{a, b\}$
State: $s(t) \in \{S_0, S_1\}$
Initial state: $s(0) = S_0$

Functions: The transition and output functions are

$$s(t+1) = \begin{cases} S_0 & \text{if } (s(t) = S_0 \\ & \quad \text{and } [x(t) = 0 \text{ or } x(t) = 2]) \\ & \quad \text{or } (s(t) = S_1 \text{ and } x(t) = 3) \\ S_1 & \text{otherwise} \end{cases}$$

$$z(t) = \begin{cases} a & \text{if } s(t) = S_0 \\ b & \text{if } s(t) = S_1 \end{cases}$$

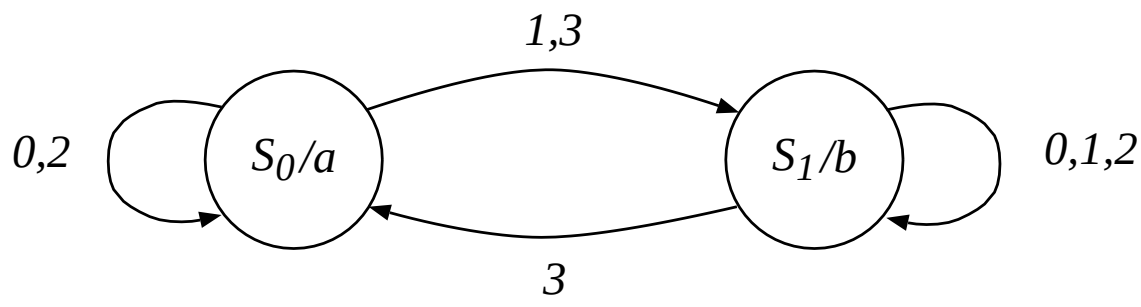


Figure 7.7: STATE DIAGRAM FOR EXAMPLE 7.7

STATE NAMES

Example 7.8: INTEGERS AS STATE NAMES

A MODULO-64 COUNTER:- A modulo-64 counter counts modulo-64 of the number of 1's in the input sequence.

Input: $x(t) \in \{0, 1\}$
Output: $z(t) \in \{0, 1, 2, \ldots, 63\}$
State: $s(t) \in \{0, 1, 2, \ldots, 63\}$
Initial state: $s(0) = 0$

Functions: The transition and output functions are

$$s(t+1) = [s(t) + x(t)] \bmod 64$$
$$z(t) = s(t)$$

Example 7.9: VECTORS AS STATE NAMES

Consider a sequential system that records (counts) the occurrence of 55 different events, which take place one at a time. When count of event i becomes a multiple of 100, the output z(t)=i, otherwise z(t)=0.

Input: $e(t) \in \{1, 2, \ldots, 55\}$

Output: $z(t) \in \{0, 1, 2, \ldots, 55\}$

State: $\underline{s}(t) = (s_{55}, \ldots, s_1), \quad s_i \in \{0, 1, 2, \ldots, 99\}$

Initial state: $\underline{s}(0) = (0, 0, \ldots, 0)$

Functions: The transition and output functions are

$$s_i(t+1) = \begin{cases} [s_i(t) + 1] \bmod 100 & \text{if } e(t) = i \\ & i = 1, 2, \ldots, 55 \\ s_i(t) & \text{otherwise} \end{cases}$$

$$z(t) = \begin{cases} i & \text{if } e(t) = i \text{ and } s_i(t) = 99 \\ 0 & \text{otherwise} \end{cases}$$

TIME BEHAVIOR AND FINITE-STATE MACHINES

- STATE DESCRIPTION ⇒ I/O SEQUENCE (Example 7.10)

Initial state: $s(0) = S_2$

Functions: Transition and output functions are

PS	$x(t)$			
	a	b	c	
S_0	S_0	S_1	S_1	p
S_1	S_2	S_0	S_1	q
S_2	S_2	S_3	S_0	q
S_3	S_0	S_1	S_2	p
	NS			$z(t)$

t	0	1	2	3	4
x	a	b	c	a	
s	S_2	S_2	S_3	S_2	S_2
z	q	q	p	q	

TIME BEHAVIOR ⇒ STATE DESCRIPTION

- NOT ALL TIME-BEHAVIORS ARE REALIZABLE BY FINITE STATE MACHINE :

$$z(t) = \begin{cases} 1 & \text{if } x(0, t) \text{ has same number of } 0\text{'s and } 1\text{'s} \\ 0 & \text{otherwise} \end{cases}$$

$s(t)$ = DIFFERENCE BETWEEN NUMBER OF 1'S AND 0'S

$$s(t+1) = \begin{cases} s(t) + 1 & \text{if } x(t) = 1 \\ s(t) - 1 & \text{otherwise} \end{cases}$$

$$z(t) = \begin{cases} 1 & \text{if } s(t) = 0 \\ 0 & \text{otherwise} \end{cases}$$

⇒ DIFFERENCE UNBOUNDED: NOT A FINITE-STATE SYSTEM

PROCEDURE FOR OBTAINING FSM FROM TIME BEHAVIOR

1. DETERMINE A SET OF STATES
2. DETERMINE THE TRANSITION FUNCTION
3. DETERMINE THE OUTPUT FUNCTION

- Example 7.11

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if} \quad x(t-3, t) = 1101 \\ 0 & \text{otherwise} \end{cases}$

- PATTERN DETECTOR $\Rightarrow$ DETECT SUBPATTERNS

- Example 7.11

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if} \quad x(t-3, t) = 1101 \\ 0 & \text{otherwise} \end{cases}$

- PATTERN DETECTOR $\Rightarrow$ DETECT SUBPATTERNS

State	indicates that
S_{init}	Initial state; also no subpattern
S_1	First symbol (1) of pattern has been detected
S_{11}	Subpattern 11 has been detected
S_{110}	Subpattern 110 has been detected

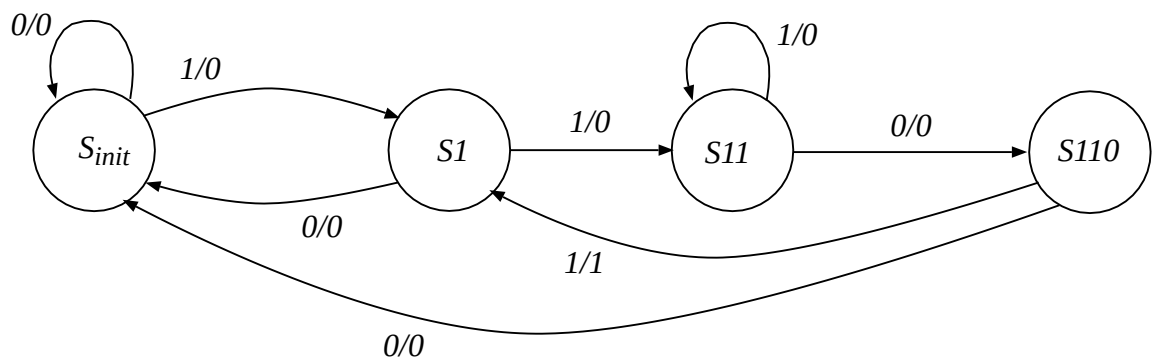


Figure 7.8: STATE DIAGRAM FOR Example 7.11

FINITE-MEMORY SEQUENTIAL SYSTEMS

A sequential system has finite memory of length m if z(t) depends only on last m input values.

$$z(t) = F(x(t-m+1, t))$$

Example 7.12:

$$z(t) = \begin{cases} p & \text{if} \quad x(t-3, t) = aaba \\ q & \text{otherwise} \end{cases}$$

⇒ FINITE MEMORY OF LENGTH FOUR

- ALL FINITE-MEMORY MACHINES ARE FS SYSTEMS
- NOT ALL FS SYSTEMS ARE FINITE MEMORY

$$z(t) = \begin{cases} 1 & \text{if} \quad \text{number of } 1\text{'s in } x(0, t) \text{ is even} \\ 0 & \text{otherwise} \end{cases}$$

EQUIVALENT SEQUENTIAL SYSTEMS: SAME TIME BEHAVIOR

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if} \quad x(t-2, t) = 101 \\ 0 & \text{otherwise} \end{cases}$

t	0	1	2	3	4	5	6	7	8
x	0	0	1	0	1	0	1	0	0
z	0	0	0	0	1	0	1	0	0

EQUIVALENT SEQUENTIAL SYSTEMS: SAME TIME BEHAVIOR

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if} \quad x(t-2, t) = 101 \\ 0 & \text{otherwise} \end{cases}$

t	0	1	2	3	4	5	6	7	8
x	0	0	1	0	1	0	1	0	0
z	0	0	0	0	1	0	1	0	0

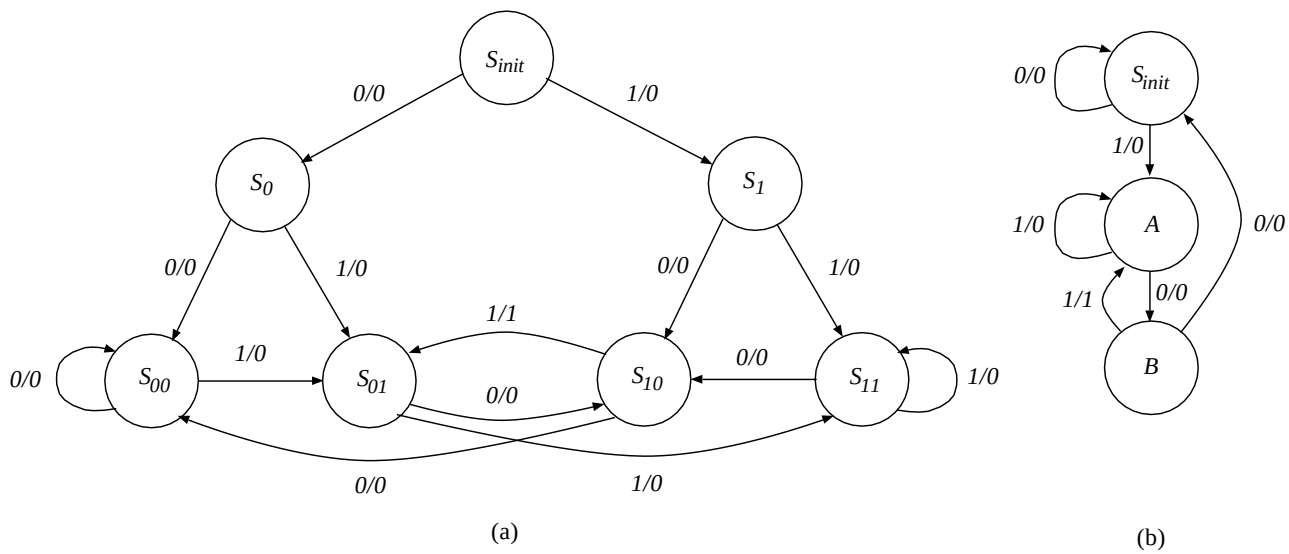


Figure 7.13: a) STATE DIAGRAM WITH REDUNDANT STATES; b) REDUCED STATE DIAGRAM

EQUIVALENT SEQUENTIAL SYSTEMS: SAME TIME BEHAVIOR

- Two sequential systems are equivalent if they have same time behaviour.
- There can be several state descriptions for same time behaviour.
- In many instances, simplest description corresponds to the description with minimum number of states.
- However, the description with minimum number of states is not always the simplest nor does it always leads to the best implementation.

DISTINGUISHABLE STATES

- Two states S_V and S_W of sequential system are distinguishable if there exists at least one finite input that generates different output sequences, when the input sequence is applied to the system in state S_V or S_W.
- A sequence that distinguishes these states is called a distinguishing sequence of (S_V ,S_W).
- If there exists a distinguishing sequence of length k for pair (S_V ,S_W), then (S_V ,S_W) are k-distinguishable. In mathematical notation, S_V and S_W are k-distinguishable if there exists an input sequence x(t,t+k-1), such that

$$z(x(t, t+k-1), S_v) \neq z(x(t, t+k-1), S_w)$$

- Where z(x(t_1, t_2),S) is the output sequence when the input sequence x(t_1, t_2) is applied with S being the state at time t_1.

EQUIVALENT STATES

- Two states that are not k-distinguishable are said to be k-equivalent.
- The partition of states into the k-equivalent classes is called P_k.
- Two states that are not distinguishable for any k are said to be equivalent.
- The partition into equivalent states is called P.

Example 7.14

Input: $x(t) \in \{a, b, c\}$
Output: $z(t) \in \{0, 1\}$
State: $s(t) \in \{A, B, C, D, E, F\}$
Initial state: $s(0) = A$

Functions: TRANSITION AND OUTPUT

PS	$x = a$	$x = b$	$x = c$
A	$E, 0$	$D, 1$	$B, 0$
B	$F, 0$	$D, 0$	$A, 1$
C	$E, 0$	$B, 1$	$D, 0$
D	$F, 0$	$B, 0$	$C, 1$
E	$C, 0$	$F, 1$	$F, 0$
F	$B, 0$	$C, 0$	$F, 1$
	NS, z		

Example 7.14 (cont.)

- A and B ARE 1-DISTINGUISHABLE BECAUSE

$$z(b, A) \neq z(b, B)$$

- A and C ARE 1-EQUIVALENT BECAUSE

$$z(x(t), A) = z(x(t), C), \quad for\ all\ x(t) \in I$$

- A and C ARE ALSO 2-EQUIVALENT BECAUSE

$$\begin{aligned}
z(aa, A) &= z(aa, C) = 00 \\
z(ab, A) &= z(ab, C) = 01 \\
z(ac, A) &= z(ac, C) = 00 \\
z(ba, A) &= z(ba, C) = 10 \\
z(bb, A) &= z(bb, c) = 10 \\
z(bc, A) &= z(bc, C) = 11 \\
z(ca, A) &= z(ca, C) = 00 \\
z(cb, A) &= z(cb, C) = 00 \\
z(cc, A) &= z(cc, C) = 01
\end{aligned}$$

PROCEDURE: SUMMARY

1. OBTAIN P_1 (look at the outputs)

2. OBTAIN P_{i+1} FROM P_i
   BY GROUPING STATES THAT ARE i-EQUIVALENT
   AND WHOSE CORRESPONDING SUCCESSORS
   ARE i-EQUIVALENT

3. TERMINATE WHEN $P_{i+1} = P_i$

4. WRITE THE REDUCED TABLE

Example 7.14

PS	$x = a$	$x = b$	$x = c$
A	$E, 0$	$D, 1$	$B, 0$
B	$F, 0$	$D, 0$	$A, 1$
C	$E, 0$	$B, 1$	$D, 0$
D	$F, 0$	$B, 0$	$C, 1$
E	$C, 0$	$F, 1$	$F, 0$
F	$B, 0$	$C, 0$	$F, 1$
	NS, z		

P1={A, C, E}, {B, D, F}

P2={A, C, E}, {B, D}, {F}

P3={A, C}, {E}, {B, D}, {F}

P4={A, C}, {E}, {B, D}, {F}

Example 7.15 (cont.)

THE MINIMAL SYSTEM:

PS	$x = a$	$x = b$	$x = c$
A	$E, 0$	$B, 1$	$B, 0$
B	$F, 0$	$B, 0$	$A, 1$
E	$A, 0$	$F, 1$	$F, 0$
F	$B, 0$	$A, 0$	$F, 1$
	NS, z		

	INPUT	
PS	**X=0**	**X=1**
A	F, 0	B, 0
B	D, 0	C, 0
C	F, 0	E, 0
D	G, 1	A, 0
E	D, 0	C, 0
F	F, 1	B, 1
G	G, 0	H, 1
H	G, 1	A, 0

	INPUT	
PS	**X=0**	**X=1**
A	F, 0	B, 0
B	D, 0	C, 0
C	F, 0	E, 0
D	G, 1	A, 0
E	D, 0	C, 0
F	F, 1	B, 1
G	G, 0	H, 1
H	G, 1	A, 0

P1={A, B, C, E}, {D, H}, {F}, {G}

P2={A, C}, {B, E}, {D, H}, {F}, {G}

P3={A, C}, {B, E}, {D, H}, {F}, {G}

	INPUT	
PS	**X=0**	**X=1**
A	B, 1	D,1
B	C, 0	F, 0
C	B, 1	D,1
D	F, 0	E, 0
E	B, 1	D,1
F	F, 0	F, 0

P1={A, C, E}, {B, D, F}

P2={A, C, E}, {B}, {D}, { F}

P3={A, C, E}, {B}, {D}, { F}

SPECIFICATION OF SEQUENTIAL SYSTEMS

- Problem statement
- High-level specification (Input, output, function)
- State diagram
- State description
- Binary-level specification

BINARY SPECIFICATION OF SEQUENTIAL SYSTEMS

- THE STATE CODING IS CALLED *STATE ASSIGNMENT*
- CODING FUNCTIONS:

Input $C_I : I \rightarrow \{0, 1\}^n$
Output $C_O : O \rightarrow \{0, 1\}^m$
State $C_S : S \rightarrow \{0, 1\}^k$

Example 7.16

PS	$x = a$	$x = b$	$x = c$
A	$E, 0$	$B, 1$	$B, 0$
B	$F, 0$	$B, 0$	$A, 1$
E	$A, 0$	$F, 1$	$F, 0$
F	$B, 0$	$A, 0$	$F, 1$
	NS, z		

BINARY CODING

Input code

$x(t)$	$x_1(t)x_0(t)$
a	00
b	01
c	10

Output code

$z(t)$	
0	0
1	1

State assignment

$s(t)$	$s_1(t)s_0(t)$
A	00
B	01
E	10
F	11

- THE RESULTING BINARY SPECIFICATION:

$s_1(t)s_0(t)$	$x_1x_0 = 00$	$x_1x_0 = 01$	$x_1x_0 = 10$
00	10, 0	01, 1	01, 0
01	11, 0	01, 0	00, 1
10	00, 0	11, 1	11, 0
11	01, 0	00, 0	11, 1
	$s_1(t+1)s_0(t+1), z$		

SPECIFICATION OF DIFFERENT TYPES OF SEQUENTIAL SYSTEMS

MODULO-p COUNTER: 0, 1, 2, ..., p-1, 0, 1, ...

Input: *x(t)={0, 1}*

Output *z(t)=(0, 1, 2,.....,p-1)*

$$z(t) = [\sum_{i=0}^{t} x(i)] \; mod \; p$$
$$s(t+1) = [s(t) + x(t)] \; mod \; p$$
$$z(t) = s(t) \quad \text{(if same coding)}$$

Input: $x(t) \in \{0, 1\}$
Output: $z(t) \in \{0, 1, 2, \ldots, p\text{-}1\}$
State: $s(t) \in \{0, 1, 2, \ldots, p\text{-}1\}$
Initial state: $s(0) = 0$

SPECIFICATION OF DIFFERENT TYPES OF SEQUENTIAL SYSTEMS

MODULO-p COUNTER: 0, 1, 2, ..., p-1, 0, 1, ...

$$
\begin{aligned}
z(t) &= [\sum_{i=0}^{t} x(i)] \; mod \; p \\
s(t+1) &= [s(t) + x(t)] \; mod \; p \\
z(t) &= s(t) \quad \text{(if same coding)}
\end{aligned}
$$

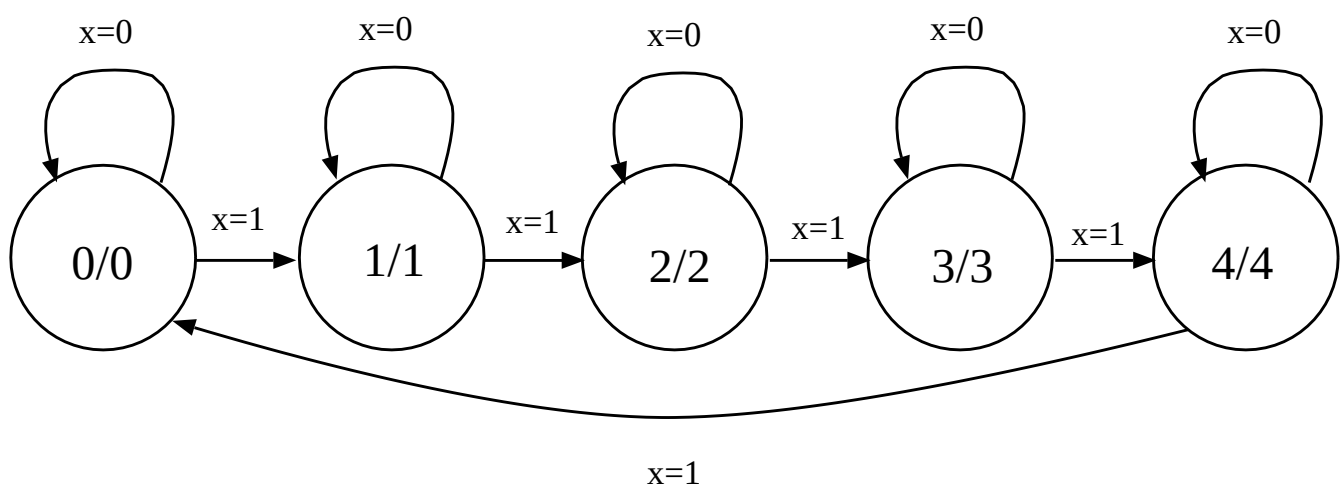


Figure 7.17: STATE DIAGRAM OF A MODULO-5 COUNTER

Pattern recognizer: recognizes the pattern P=(p0,p1,p2,……,pm-1)

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

Function: $z(t) = \begin{cases} 1 & \text{if} \quad x(t - m+1, t) = P \\ 0 & \text{otherwise} \end{cases}$

There are two possible state descriptions for pattern recognizers

1) Uses a vector to represent state

2) Uses states that corresponds to partial recognition of pattern

vector to represent state

Sequential system that recognizes the pattern 0101.

Input: $x(t) \in \{0, 1\}$

Output: $z(t) \in \{0, 1\}$

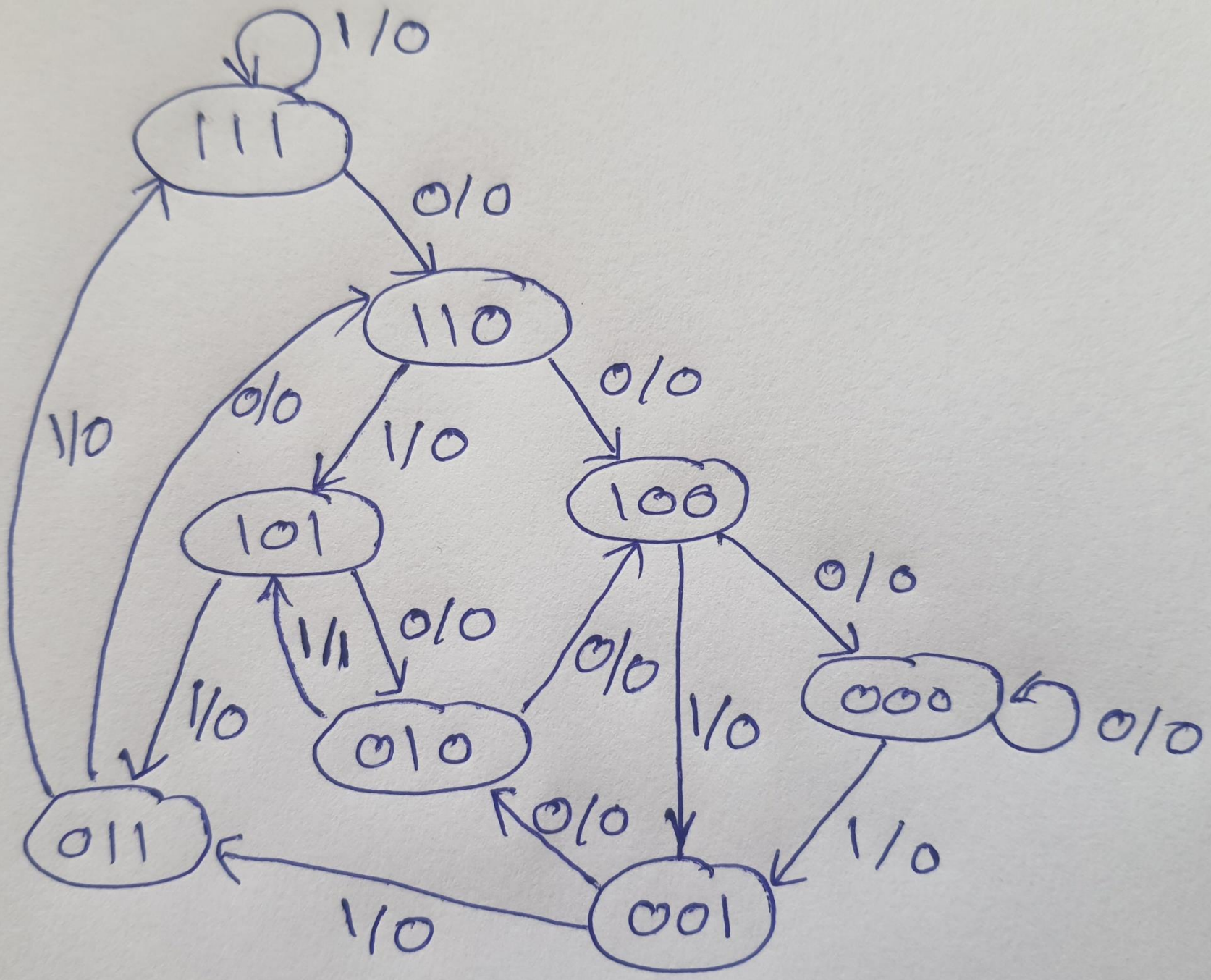
State: $s(t) \in (S_2, S_1, S_0)$ *and* $S_i=\{0,1\}$

Initial state: $s(0) =\{1,1,1\}$

$S_i(t+1)=S_{i-1}(t)$ *if* $i=\{1,2\}$

$S_0(t+1)=x(t)$

$$z(t)= \begin{cases} 1 & \text{if } (s(t),x(t))=0101 \\ 0 & \text{otherwise} \end{cases}$$


111
110
101
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010
000
011
001
1/0
0/0
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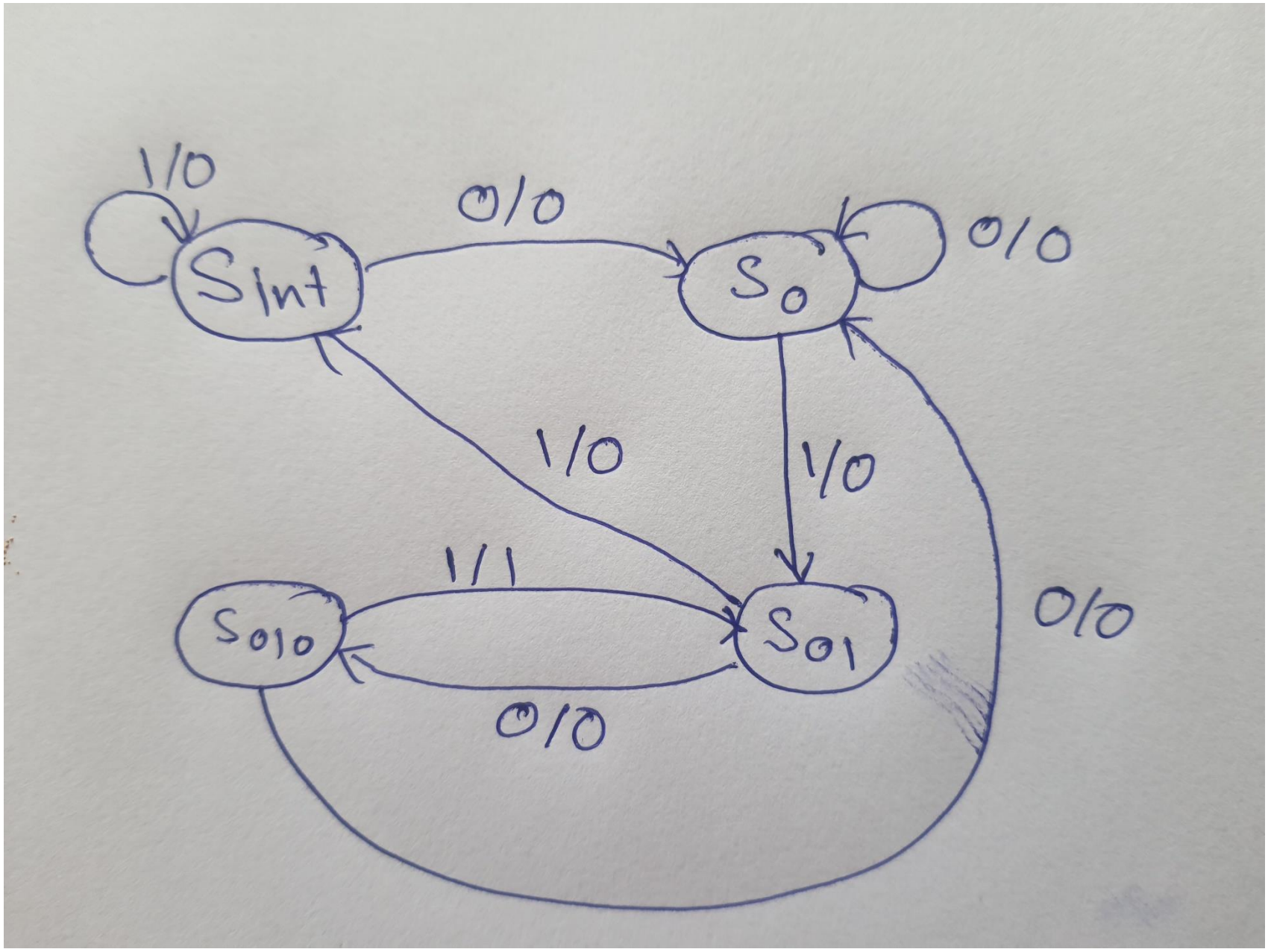
0/0
1/0

Uses states that corresponds to partial recognition of pattern



CONTROLLERS

- THE STATE DESCRIPTION IS PRIMARY
- FSM PRODUCING CONTROL SIGNALS
- CONTROL SIGNALS DETERMINE ACTIONS PERFORMED IN OTHER PARTS OF SYSTEM
- *AUTONOMOUS*: FIXED SEQUENCE OF STATES, INDEPENDENT OF INPUTS

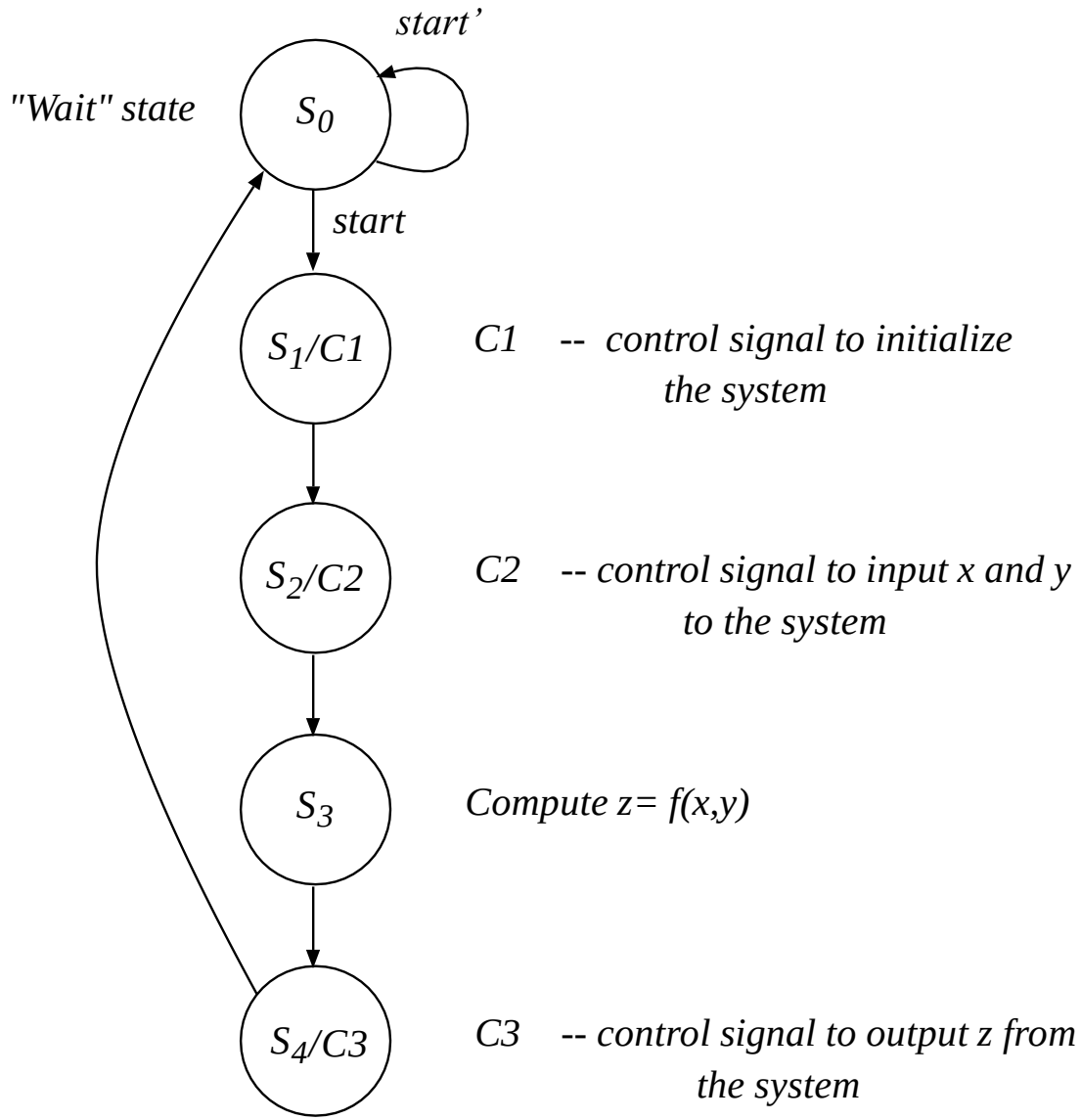


Figure 7.10: AUTONOMOUS CONTROLLER: STATE DIAGRAM.

AUTONOMOUS CONTROLLER

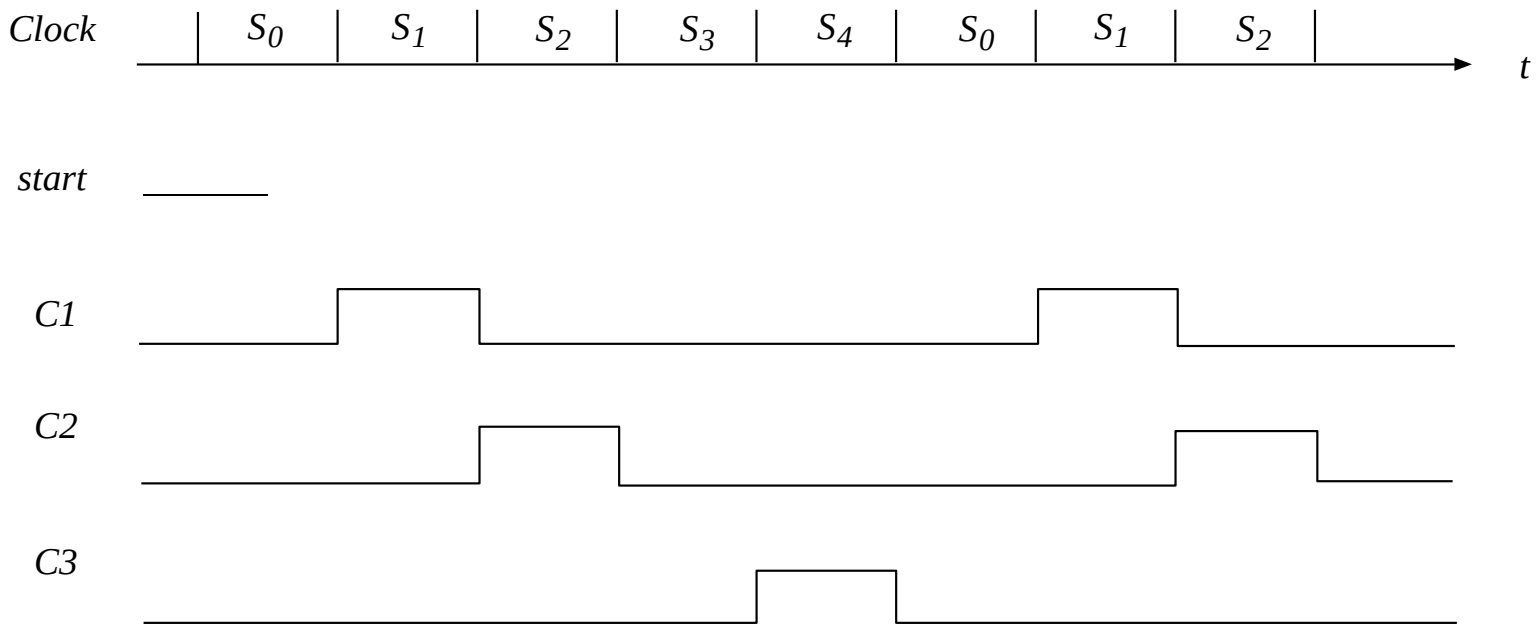
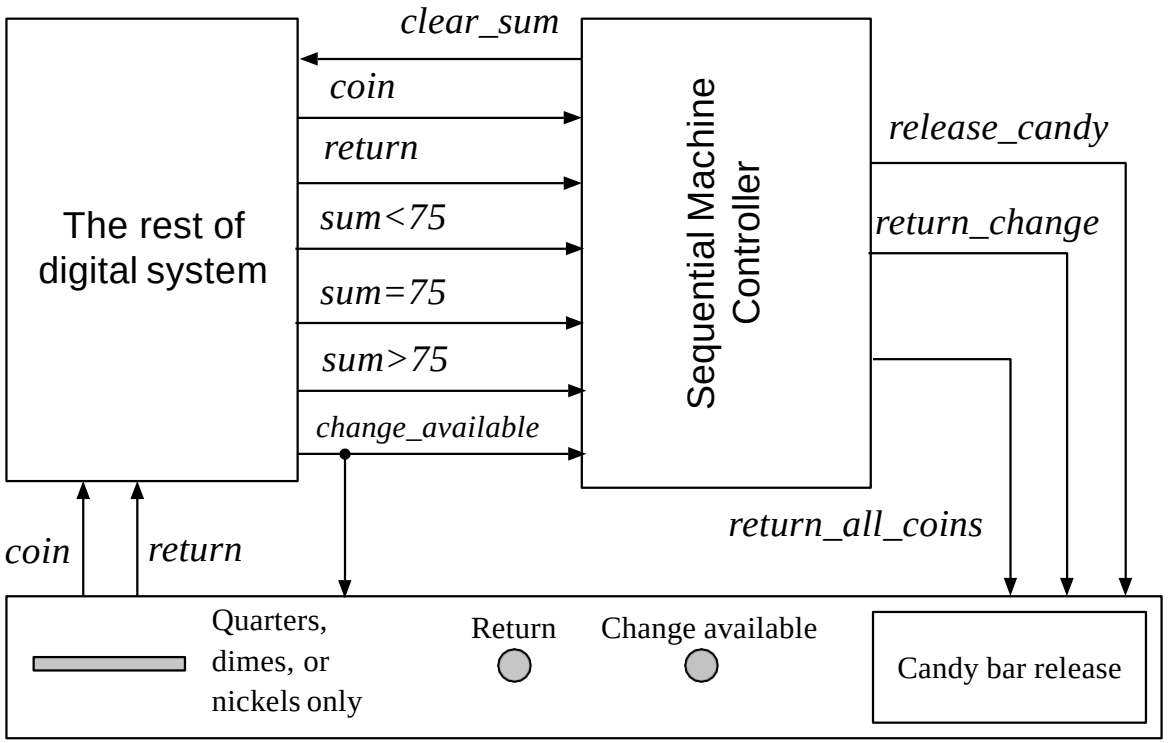


Figure 7.9: AUTONOMOUS CONTROLLER: TIMING DIAGRAM.

GENERAL CONTROLLER



Note: *coin*• *return* = *0*

Figure 7.11: CONTROLLER FOR SIMPLE VENDING MACHINE: BLOCK DIAGRAM.

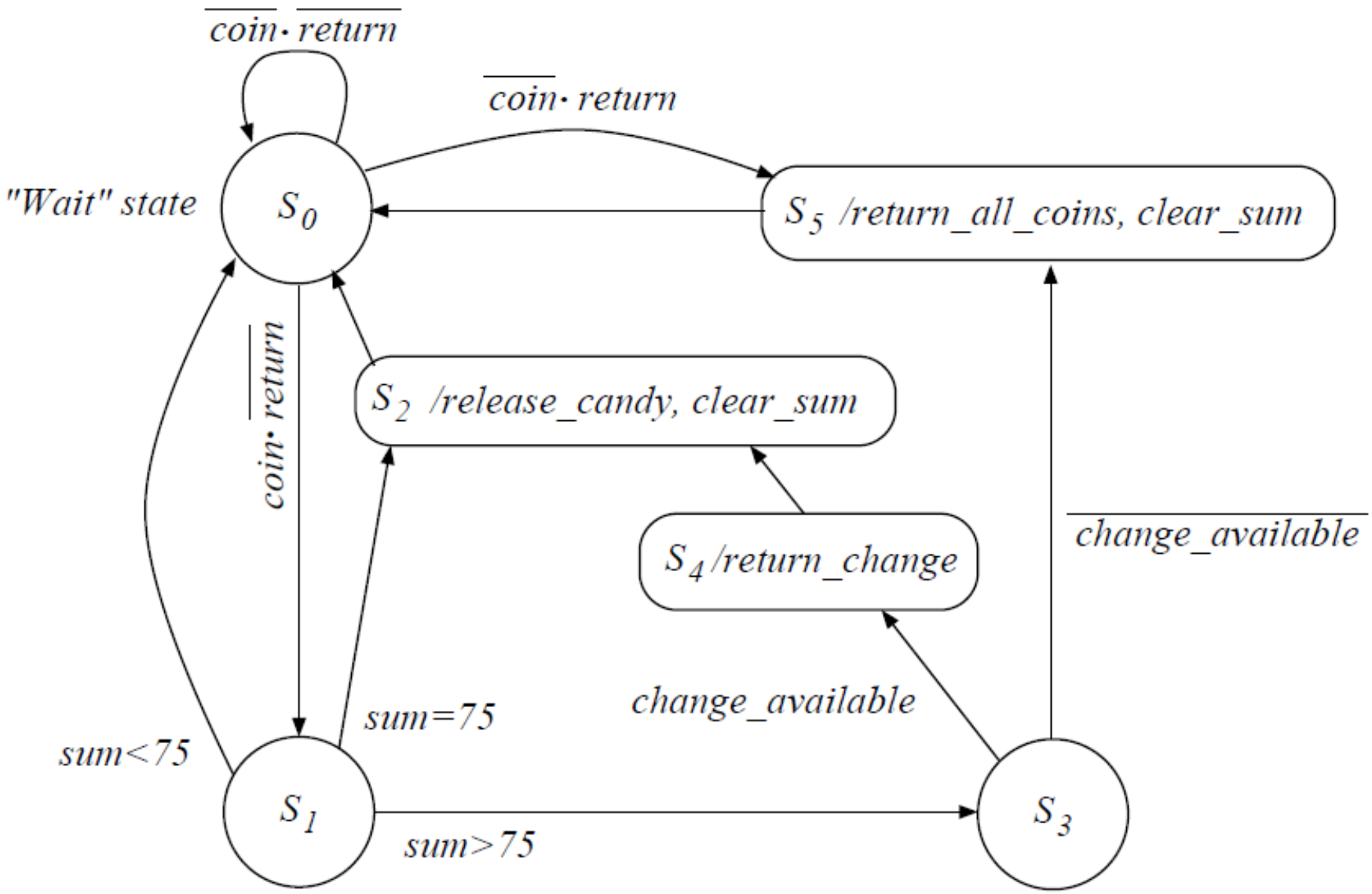


Figure 7.12: CONTROLLER FOR SIMPLE VENDING MACHINE: STATE DIAGRAM.